

10.5 The Riesz Representation Theorem

Chapter 4 built the dual space V^* of all linear functionals on V , proved that $\dim V^* = \dim V$ in finite dimension, and then delivered a warning: the resulting isomorphism $V \cong V^*$ is not canonical. Every basis of V gives one, different bases give different ones, and no argument singles out a preferred choice (Corollary 4.1.4, and the contrast with the natural map $\text{ev}_V: V \rightarrow V^{**}$ of Theorem 4.4.6). The chapter closed by promising that an inner product would supply the missing ingredient. This section keeps the promise.

The mechanism is simple to state. Fix $\mathbf{w} \in V$ and let $\varphi_{\mathbf{w}}(\mathbf{v}) = \langle \mathbf{v}, \mathbf{w} \rangle$. Because the inner product is linear in the **first** slot, $\varphi_{\mathbf{w}}$ is a linear functional. The Riesz Representation Theorem says that in finite dimension there are no others.

Throughout, V is a finite-dimensional inner product space over $F = \mathbb{R}$ or $F = \mathbb{C}$, unless the text says otherwise.

10.5.1 Every functional is an inner product

In F^n , Theorem 4.1.1 already told us that every functional has the form $\mathbf{x} \mapsto a_1x_1 + \cdots + a_nx_n$ for a unique row of coefficients. With the standard inner product $\langle \mathbf{x}, \mathbf{y} \rangle = \sum_i x_i \overline{y_i}$, that reads $\varphi(\mathbf{x}) = \langle \mathbf{x}, \mathbf{w} \rangle$ with $\mathbf{w} = (\overline{a_1}, \dots, \overline{a_n})$. So in coordinates the theorem is a restatement of a fact we already have; the content is that it survives the removal of coordinates.

Theorem 10.5.1 — Riesz Representation Theorem

Let V be a finite-dimensional inner product space over F and let $\varphi \in V^*$. Then there is a **unique** $\mathbf{w} \in V$ such that

$$\varphi(\mathbf{v}) = \langle \mathbf{v}, \mathbf{w} \rangle \quad \text{for every } \mathbf{v} \in V.$$

If $(\mathbf{e}_1, \dots, \mathbf{e}_n)$ is any orthonormal basis of V , then $\mathbf{w} = \sum_{i=1}^n \overline{\varphi(\mathbf{e}_i)} \mathbf{e}_i$.

Idea. Fix an orthonormal basis. Then a linear functional is determined by the n numbers $\varphi(\mathbf{e}_i)$, and a vector is determined by its n coordinates, which Theorem 10.2.1 reads off as inner products. Matching the two lists tells us what $\mathbf{w}$ must be, and the only care needed is the conjugate: $\mathbf{w}$ sits in the second slot, so its coordinates come out conjugated. Uniqueness is the standard move — if two vectors pair identically with everything, subtract and pair the difference with itself.

Proof. Existence. Let $(\mathbf{e}_1, \dots, \mathbf{e}_n)$ be an orthonormal basis of V , which exists by Theorem 10.2.2, and put

$$\mathbf{w} = \sum_{i=1}^n \overline{\varphi(\mathbf{e}_i)} \mathbf{e}_i.$$

For each j , conjugate-linearity in the second slot and $\langle \mathbf{e}_j, \mathbf{e}_i \rangle = \delta_{ij}$ give

$$\langle \mathbf{e}_j, \mathbf{w} \rangle = \sum_{i=1}^n \overline{\overline{\varphi(\mathbf{e}_i)}} \langle \mathbf{e}_j, \mathbf{e}_i \rangle = \varphi(\mathbf{e}_j).$$

So the two linear functionals $\mathbf{v} \mapsto \langle \mathbf{v}, \mathbf{w} \rangle$ and φ agree on a basis, hence are equal by Theorem 3.1.5. (The first is linear because the inner product is linear in the first slot.)

Uniqueness. Suppose $\langle \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{v}, \mathbf{w}' \rangle$ for every $\mathbf{v}$. Then $\langle \mathbf{v}, \mathbf{w} - \mathbf{w}' \rangle = 0$ for every $\mathbf{v}$, by additivity in the second slot. Taking $\mathbf{v} = \mathbf{w} - \mathbf{w}'$ gives $\|\mathbf{w} - \mathbf{w}'\|^2 = 0$, so $\mathbf{w} = \mathbf{w}'$ by positive definiteness. ■

Call $\mathbf{w}$ the **Riesz vector** of φ . The formula is worth reading twice: the coordinates of $\mathbf{w}$ in an orthonormal basis are the conjugates of the values of φ on that basis. Over $\mathbb{R}$ the conjugation is invisible, which is why real textbooks can omit it and complex ones cannot.

Example. (Two Riesz vectors)

- (a) On $V = \mathbb{R}[x]_{\leq 1}$ with $\langle p, q \rangle = \int_0^1 p(t)q(t) dt$, find the Riesz vector of $\varphi(p) = p(0)$.
- (b) On $V = M_n(\mathbb{C})$ with the Frobenius inner product $\langle \mathbf{A}, \mathbf{B} \rangle = \text{tr}(\mathbf{B}^* \mathbf{A})$, find the Riesz vector of $\varphi(\mathbf{A}) = \text{tr } \mathbf{A}$.

Solution.

- (a) We look for $w = a + bx$ with $\langle p, w \rangle = p(0)$ for all p . By linearity it is enough to impose this on the basis $(1, x)$:

$$\int_0^1 (a + bt) dt = a + \frac{b}{2} = 1, \quad \int_0^1 t(a + bt) dt = \frac{a}{2} + \frac{b}{3} = 0.$$

The second equation gives $a = -\frac{2}{3}b$; substituting into the first gives $-\frac{2}{3}b + \frac{1}{2}b = -\frac{1}{6}b = 1$, so $b = -6$ and $a = 4$. Hence $w = 4 - 6x$.

Check. $\int_0^1 (4 - 6t) dt = 4 - 3 = 1$ and $\int_0^1 t(4 - 6t) dt = 2 - 2 = 0$, as required; and for a general $p = c + dx$ the pairing gives $c \cdot 1 + d \cdot 0 = c = p(0)$. Note how little w resembles “evaluation at 0”: the Riesz vector is manufactured by the inner product, not by the point 0.

- (b) We look for $\mathbf{W}$ with $\text{tr}(\mathbf{W}^* \mathbf{A}) = \text{tr } \mathbf{A}$ for every $\mathbf{A}$. Taking $\mathbf{W} = \mathbf{I}_n$ works, since $\mathbf{I}_n^* = \mathbf{I}_n$ and $\text{tr}(\mathbf{I}_n \mathbf{A}) = \text{tr } \mathbf{A}$. By uniqueness, $\mathbf{W} = \mathbf{I}_n$ is the answer. (This example was not solved by the basis formula, but by guessing and then invoking uniqueness — a legitimate and often faster route.)

10.5.2 The Riesz map, and the data Chapter 4 was missing

Uniqueness in Theorem 10.5.1 lets us read the theorem as a statement about a map.

Corollary 10.5.2 — The Riesz map

Let V be a finite-dimensional inner product space over F . Define

$$\Phi: V \rightarrow V^*, \quad \Phi(\mathbf{w}) = \langle \cdot, \mathbf{w} \rangle.$$

Then Φ is a bijection, and for all $\mathbf{w}, \mathbf{w}' \in V$ and $c \in F$,

$$\Phi(\mathbf{w} + \mathbf{w}') = \Phi(\mathbf{w}) + \Phi(\mathbf{w}'), \quad \Phi(c\mathbf{w}) = \bar{c}\Phi(\mathbf{w}).$$

In particular Φ is an isomorphism of vector spaces when $F = \mathbb{R}$, and is **conjugate-linear** when $F = \mathbb{C}$.

Proof. Each $\Phi(\mathbf{w})$ lies in V^* because the inner product is linear in the first slot. Surjectivity is the existence half of [Theorem 10.5.1](#) and injectivity is the uniqueness half (if $\Phi(\mathbf{w}) = \Phi(\mathbf{w}')$, then $\mathbf{w} = \mathbf{w}'$). For the two rules, evaluate at an arbitrary $\mathbf{v}$: additivity in the second slot gives $\langle \mathbf{v}, \mathbf{w} + \mathbf{w}' \rangle = \langle \mathbf{v}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w}' \rangle$, and conjugate-homogeneity in the second slot gives $\langle \mathbf{v}, c\mathbf{w} \rangle = \bar{c}\langle \mathbf{v}, \mathbf{w} \rangle$. Over $\mathbb{R}$ we have $\bar{c} = c$, so Φ is linear, and a linear bijection is an isomorphism (Theorem 0.4.7). ■

This is the payoff Chapter 4 promised. There, $V \cong V^*$ was proved by choosing a basis $\mathcal{B}$ and sending $\mathbf{v}_i \mapsto \varphi_i$, and the warning was that the isomorphism changes when $\mathcal{B}$ does: a vector does not come with “its” functional. Here Φ is written down with no basis at all. It is not free of choices — it depends on $\langle \cdot, \cdot \rangle$ — but the choice has been made once, in advance, and it is a choice of *geometry* rather than of coordinates. **The inner product is exactly the extra data that turns $V \cong V^*$ from a counting fact into a formula.**

The two descriptions fit together neatly. Let $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be an **orthonormal** basis of V , with dual basis $(\varphi_1, \dots, \varphi_n)$ (Theorem 4.1.3). Then $\Phi(\mathbf{v}_i)$ is the functional $\mathbf{v} \mapsto \langle \mathbf{v}, \mathbf{v}_i \rangle$, which sends $\mathbf{v}_j$ to δ_{ij} ; that is,

$$\Phi(\mathbf{v}_i) = \varphi_i \quad (i = 1, \dots, n),$$

over either field. Over $\mathbb{R}$, where Φ is linear, this says that Φ is Chapter 4’s basis-dependent isomorphism $\mathbf{v}_i \mapsto \varphi_i$, for the inner product that declares $\mathcal{B}$ orthonormal. Choosing a basis and choosing an inner product are two ways of supplying the same missing data.

Quick check. Let $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be an orthonormal basis of a **complex** inner product space and let $(\varphi_1, \dots, \varphi_n)$ be its dual basis. What is the Riesz vector of $i\varphi_1$?

Solution. By the identity $\Phi(\mathbf{v}_i) = \varphi_i$ above, which holds over $\mathbb{C}$ as well, φ_1 has Riesz vector $\mathbf{v}_1$. For $i\varphi_1$, use conjugate-homogeneity in reverse: $\Phi(-i\mathbf{v}_1) = \overline{(-i)}\Phi(\mathbf{v}_1) = i\varphi_1$. So the Riesz vector of $i\varphi_1$ is $-i\mathbf{v}_1$, not $i\mathbf{v}_1$. Alternatively, apply the formula of [Theorem 10.5.1](#): the coordinates of the Riesz vector are $\overline{(i\varphi_1)(\mathbf{v}_j)} = \overline{i\delta_{1j}} = -i\delta_{1j}$.

Warning. Over $\mathbb{C}$, the Riesz map is not linear, so it is not an isomorphism of complex vector spaces. It is a bijection, and it is linear over $\mathbb{R}$, but $\Phi(i\mathbf{w}) = -i\Phi(\mathbf{w})$. Complex spaces V and V^* are still isomorphic — they have the same dimension — but the isomorphism produced by an inner product is the conjugate-linear Φ . Most of what we do with Φ (Section 6 builds the adjoint from it) is unaffected, because the conjugation cancels when Φ is used twice.

Warning. The Riesz vector depends on the inner product, not on φ alone. On $\mathbb{R}^2$ let $\varphi(\mathbf{x}) = x_1 + x_2$. With the dot product, $\varphi(\mathbf{x}) = \langle \mathbf{x}, (1, 1) \rangle$, so $\mathbf{w} = (1, 1)$. With the weighted inner product $\langle \mathbf{x}, \mathbf{y} \rangle = 2x_1y_1 + x_2y_2$, we need $2w_1 = 1$ and $w_2 = 1$, so $\mathbf{w} = (\frac{1}{2}, 1)$. Same functional, different representing vector. Whenever two inner products are in play, the notation must say which Φ is meant.

10.5.3 Annihilators become orthogonal complements

Section 3 warned that the annihilator $U^0 \subseteq V^*$ and the orthogonal complement $U^\perp \subseteq V$ are different objects that happen to have the same dimension. The Riesz map explains the coincidence: it carries one onto the other.

Theorem 10.5.3 — Perp corresponds to the annihilator

Let V be a finite-dimensional inner product space over F , let U be a subspace and let Φ be the Riesz map. Then

$$\Phi(U^\perp) = U^0.$$

Equivalently, a functional $\varphi \in V^*$ annihilates U if and only if its Riesz vector lies in $U^\perp$.

Proof. Let $\mathbf{w} \in V$. Then

$$\Phi(\mathbf{w}) \in U^0 \iff \langle \mathbf{u}, \mathbf{w} \rangle = 0 \text{ for every } \mathbf{u} \in U \iff \mathbf{w} \in U^\perp,$$

the first equivalence being the definition of the annihilator (Definition 4.2.1) applied to the functional $\Phi(\mathbf{w}) = \langle \cdot, \mathbf{w} \rangle$, and the second the definition of the orthogonal complement (Definition 10.3.1). Since Φ is a bijection (Corollary 10.5.2), the set $\Phi(U^\perp)$ is therefore exactly U^0 . ■

The dimension count now has a reason rather than a coincidence: $\dim U^0 = \dim V - \dim U$ (Theorem 4.2.2) and $\dim U^\perp = \dim V - \dim U$ (Theorem 10.3.2 (c)) agree because the two spaces correspond under a bijection. (Over $\mathbb{C}$ the bijection is only conjugate-linear, but a conjugate-linear bijection still carries subspaces to subspaces of the same dimension, since $c\Phi(\mathbf{w}) = \Phi(\bar{c}\mathbf{w})$.)

Here is the correspondence in $\mathbb{R}^3$. Let $U = \text{span}((1, 2, 0))$ with the dot product. Then $U^\perp = \{(a, b, c) : a + 2b = 0\} = \text{span}((2, -1, 0), (0, 0, 1))$, and U^0 consists of the functionals $\mathbf{x} \mapsto \alpha x_1 + \beta x_2 + \gamma x_3$ with $\alpha + 2\beta = 0$, spanned by $2x_1 - x_2$ and x_3 . These are precisely $\Phi((2, -1, 0))$ and $\Phi((0, 0, 1))$.

Section 6 uses Theorem 10.5.3 in the same way for maps: the dual map $T' : W^* \rightarrow V^*$ of Chapter 4 and the adjoint $T^* : W \rightarrow V$ are the same map, read through the two Riesz maps.

10.5.4 What happens in infinite dimension

The proof used an orthonormal basis of the whole of V , so finite dimension was essential, and the theorem genuinely fails without it.

Example. (A functional with no Riesz vector)

Let $V = \mathbb{R}[x]$ with the coefficient inner product $\langle p, q \rangle = \sum_{k \geq 0} a_k b_k$ for $p = \sum_k a_k x^k$ and $q = \sum_k b_k x^k$ (a finite sum, and an inner product by Exercise (C2: A subspace with no orthogonal complement) (a)). Show that $\varphi(p) = p(1)$ has no Riesz vector.

Solution. Suppose $w = \sum_k b_k x^k$ satisfied $\langle p, w \rangle = p(1)$ for every $p \in V$. Taking $p = x^i$ gives

$$b_i = \langle x^i, w \rangle = \varphi(x^i) = 1^i = 1 \quad \text{for every } i \geq 0.$$

So every coefficient of w equals 1. But a polynomial has only finitely many non-zero coefficients, so no such w exists.

What fails is not “infinite dimension” as a slogan. Two things are missing at once, and the same example shows both. First, V has gaps: Exercise (C2: A subspace with no orthogonal complement) exhibits polynomials $p_m \in \ker \varphi$ with $\|p_m - 1\| = 1/\sqrt{m}$, so the vectors p_m crowd around the polynomial 1 without any limit being forced to lie in $\ker \varphi$. Second, φ is wildly discontinuous with respect to $\|\cdot\|$: along the same sequence, $\varphi(p_m) = 0$ for every m , while $\varphi(1) = 1$.

Repairing both defects — completing the space, and restricting attention to functionals φ with $|\varphi(\mathbf{v})| \leq C\|\mathbf{v}\|$ — restores the theorem. That statement is the Riesz representation theorem for Hilbert spaces, and it belongs to analysis, not to linear algebra; we record its existence and go no further. What survives here with no hypotheses at all is the easy direction: for every $\mathbf{w} \in V$, the map $\langle \cdot, \mathbf{w} \rangle$ is a linear functional. Only the converse needs finite dimension.

10.5.5 Exercises

10.5.5.1 A. Check your understanding

Exercise A1.

- State the Riesz Representation Theorem, including every hypothesis.
- Write down the formula for the Riesz vector of φ in terms of an orthonormal basis, and say where the conjugate comes from.
- True or false: over $\mathbb{C}$, the Riesz map $\Phi: V \rightarrow V^*$ is a linear isomorphism. Justify your answer.
- In what sense does an inner product answer the question Chapter 4 left open about $V \cong V^*$?
- Under Φ , which subspace of V^* corresponds to $U^\perp$?
- Give an example showing that the theorem fails in infinite dimension.

Solution.

- If V is a **finite-dimensional** inner product space over $\mathbb{R}$ or $\mathbb{C}$ and $\varphi \in V^*$, then there is a unique $\mathbf{w} \in V$ with $\varphi(\mathbf{v}) = \langle \mathbf{v}, \mathbf{w} \rangle$ for all $\mathbf{v} \in V$ ([Theorem 10.5.1](#)).
- $\mathbf{w} = \sum_i \overline{\varphi(\mathbf{e}_i)} \mathbf{e}_i$ for any orthonormal basis $(\mathbf{e}_1, \dots, \mathbf{e}_n)$. The conjugate appears because $\mathbf{w}$ occupies the second slot, where scalars come out conjugated: to make $\langle \mathbf{e}_j, \mathbf{w} \rangle$ equal $\varphi(\mathbf{e}_j)$, the coefficient of $\mathbf{e}_j$ in $\mathbf{w}$ must be $\overline{\varphi(\mathbf{e}_j)}$.
- False. Φ is a bijection, but $\Phi(c\mathbf{w}) = \bar{c}\Phi(\mathbf{w})$, so it is conjugate-linear and not $\mathbb{C}$ -linear ([Corollary 10.5.2](#)). It is linear over $\mathbb{R}$, and over $\mathbb{R}$ it is an isomorphism.
- Chapter 4 showed $V \cong V^*$ by counting dimensions, with an isomorphism depending on a chosen basis and no preferred choice available ([Corollary 4.1.4](#)). An inner product supplies a specific map $\Phi(\mathbf{w}) = \langle \cdot, \mathbf{w} \rangle$, written without reference to any

basis. The inner product is the extra data; picking an orthonormal basis and picking an inner product that makes a given basis orthonormal are the same act.

- (e) U^0 , the annihilator of U ([Theorem 10.5.3](#)).
- (f) On $\mathbb{R}[x]$ with the coefficient inner product, $\varphi(p) = p(1)$ has no Riesz vector: such a vector would need every coefficient equal to 1 ([Example \(A functional with no Riesz vector\)](#)).

10.5.5.2 B. Practice

Exercise B1: Riesz vectors for evaluation.

On $V = \mathbb{R}[x]_{\leq 2}$ use the node inner product $\langle p, q \rangle = p(-1)q(-1) + p(0)q(0) + p(1)q(1)$. Find the Riesz vectors of $\varphi(p) = p(0)$ and of $\psi(p) = p(1)$.

Solution. Since $-1, 0, 1$ are three distinct nodes and $\dim V = 3$, a polynomial of V is determined by its values there ([Theorem 2.7.3](#)). Writing w for a candidate Riesz vector of φ , the requirement $\langle p, w \rangle = p(0)$ reads

$$p(-1)w(-1) + p(0)w(0) + p(1)w(1) = p(0) \quad \text{for all } p \in V.$$

Taking for p the three Lagrange polynomials at the nodes, which have value 1 at one node and 0 at the others ([Definition 5.6.1](#)), the three equations become $w(-1) = 0$, $w(0) = 1$, $w(1) = 0$. The unique quadratic with those values is $w = 1 - x^2$.

For ψ the same argument gives $w(-1) = 0$, $w(0) = 0$, $w(1) = 1$, whose interpolant is $w = \frac{1}{2}x(x+1)$.

Check. For $p = 3 + 2x - x^2$ we get $p(-1) = 0$, $p(0) = 3$, $p(1) = 4$. Pairing with $1 - x^2$, whose values are 0, 1, 0, gives $3 = p(0)$; pairing with $\frac{1}{2}x(x+1)$, whose values are 0, 0, 1, gives $4 = p(1)$.

Exercise B2: Riesz vectors on matrices.

On $M_n(\mathbb{C})$ with the Frobenius inner product $\langle \mathbf{A}, \mathbf{B} \rangle = \text{tr}(\mathbf{B}^* \mathbf{A})$, find the Riesz vector of:

- (a) $\varphi(\mathbf{A}) = a_{12}$ (assume $n \geq 2$);
- (b) $\varphi_{\mathbf{C}}(\mathbf{A}) = \text{tr}(\mathbf{C}\mathbf{A})$, for a fixed $\mathbf{C} \in M_n(\mathbb{C})$.

Solution.

- (a) Let $\mathbf{E}_{12}$ be the matrix unit with a 1 in position $(1, 2)$ and zeros elsewhere. Then $\mathbf{E}_{12}^* = \mathbf{E}_{21}$, and $(\mathbf{E}_{21}\mathbf{A})_{kk} = \sum_j (\mathbf{E}_{21})_{kj} a_{jk}$, which is 0 unless $k = 2$, where it equals a_{12} . Hence $\text{tr}(\mathbf{E}_{12}^* \mathbf{A}) = a_{12} = \varphi(\mathbf{A})$, and by uniqueness in [Theorem 10.5.1](#) the Riesz vector is $\mathbf{E}_{12}$.
- (b) We need $\mathbf{W}$ with $\text{tr}(\mathbf{W}^* \mathbf{A}) = \text{tr}(\mathbf{C}\mathbf{A})$ for all $\mathbf{A}$. Taking $\mathbf{W}^* = \mathbf{C}$, that is $\mathbf{W} = \mathbf{C}^*$, works. By uniqueness it is the only answer. Part (a) is the case $\mathbf{C} = \mathbf{E}_{21}$, and the example in the text is the case $\mathbf{C} = \mathbf{I}_n$.

Exercise B3: The correspondence in $\mathbb{R}^4$.

In $\mathbb{R}^4$ with the dot product, let $U = \text{span}((1, 0, 1, 0), (0, 1, 0, 1))$. Find a basis of $U^\perp$ and a basis of U^0 , and verify that Φ carries the first to the second.

Solution. A vector (a, b, c, d) lies in $U^\perp$ exactly when $a + c = 0$ and $b + d = 0$ (Proposition 10.3.1, testing against the two spanning vectors). So

$$U^\perp = \text{span}((1, 0, -1, 0), (0, 1, 0, -1)),$$

of dimension $2 = 4 - 2$.

A functional $\varphi(\mathbf{x}) = \alpha x_1 + \beta x_2 + \gamma x_3 + \delta x_4$ lies in U^0 exactly when it kills both spanning vectors, that is, when $\alpha + \gamma = 0$ and $\beta + \delta = 0$. So $U^0 = \text{span}(\varphi_1, \varphi_2)$ with

$$\varphi_1(\mathbf{x}) = x_1 - x_3, \quad \varphi_2(\mathbf{x}) = x_2 - x_4.$$

Over $\mathbb{R}$, $\Phi(\mathbf{w})(\mathbf{x}) = \langle \mathbf{x}, \mathbf{w} \rangle = \sum_i x_i w_i$, so $\Phi((1, 0, -1, 0)) = \varphi_1$ and $\Phi((0, 1, 0, -1)) = \varphi_2$. Since Φ is linear over $\mathbb{R}$ and bijective, it carries the span of the two vectors onto the span of the two functionals, confirming Theorem 10.5.3.

10.5.5.3 C. Going deeper**Exercise C1: Riesz without an orthonormal basis.**

Let V be a finite-dimensional inner product space and let $\varphi \in V^*$ be non-zero. Write $K = \ker \varphi$.

- (a) Prove that $\dim K^\perp = 1$.
- (b) Let $\mathbf{u} \in \text{span } K^\perp$ with $\|\mathbf{u}\| = 1$. Prove that $\mathbf{w} = \overline{\varphi(\mathbf{u})} \mathbf{u}$ satisfies $\varphi(\mathbf{v}) = \langle \mathbf{v}, \mathbf{w} \rangle$ for all $\mathbf{v} \in V$.
- (c) Deduce Theorem 10.5.1, including the case $\varphi = 0$.

Hint for (b): split $\mathbf{v}$ using $V = K \oplus K^\perp$.

Solution.

- (a) Since $\varphi \neq 0$, its image is a non-zero subspace of the 1-dimensional space F , so $\text{im } \varphi = F$ and $\text{rank } \varphi = 1$. By Theorem 3.3.1, $\dim K = \dim V - 1$, and by Theorem 10.3.2 (c), $\dim K^\perp = \dim V - \dim K = 1$.
- (b) By Theorem 10.3.2 (a), $V = K \oplus K^\perp$, so each $\mathbf{v} \in V$ is $\mathbf{v} = \mathbf{k} + c\mathbf{u}$ with $\mathbf{k} \in K$ and $c \in F$. Then $\varphi(\mathbf{v}) = \varphi(\mathbf{k}) + c\varphi(\mathbf{u}) = c\varphi(\mathbf{u})$. On the other side,

$$\langle \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{k} + c\mathbf{u}, \overline{\varphi(\mathbf{u})} \mathbf{u} \rangle = \varphi(\mathbf{u})(\langle \mathbf{k}, \mathbf{u} \rangle + c\langle \mathbf{u}, \mathbf{u} \rangle) = \varphi(\mathbf{u})c,$$

using conjugate-homogeneity in the second slot (which conjugates $\overline{\varphi(\mathbf{u})}$ back to $\varphi(\mathbf{u})$), then $\langle \mathbf{k}, \mathbf{u} \rangle = 0$ because $\mathbf{u} \in K^\perp$ and $\mathbf{k} \in K$, and $\langle \mathbf{u}, \mathbf{u} \rangle = \|\mathbf{u}\|^2 = 1$. The two sides agree.

- (c) If $\varphi = 0$, then $\mathbf{w} = \mathbf{0}$ represents it. If $\varphi \neq 0$, part (a) supplies a unit vector $\mathbf{u}$ spanning the line $K^\perp$ — any non-zero vector of that line divided by its norm — and part (b) supplies $\mathbf{w}$. Uniqueness is unchanged: if $\langle \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{v}, \mathbf{w}' \rangle$ for all $\mathbf{v}$,

take $\mathbf{v} = \mathbf{w} - \mathbf{w}'$ to get $\|\mathbf{w} - \mathbf{w}'\|^2 = 0$. Note that this proof never mentions a basis of V ; it uses only the orthogonal decomposition along the hyperplane $\ker \varphi$.

Exercise C2: Evaluation at 0 is not represented.

Let $V = \mathbb{R}[x]$ with $\langle p, q \rangle = \int_0^1 p(t)q(t) dt$, and let $\varphi(p) = p(0)$. Suppose, for a contradiction, that $w = \sum_{j=0}^d c_j x^j$ satisfied $\langle p, w \rangle = p(0)$ for every $p \in V$.

- (a) Show that $\sum_{j=0}^d \frac{c_j}{k+j+1} = 0$ for every integer $k \geq 1$, and that the same sum equals 1 for $k = 0$.
- (b) Let $N(s) = \sum_{j=0}^d c_j \prod_{i \neq j} (s+i+1)$, the product running over $i \in \{0, \dots, d\} \setminus \{j\}$. Show that $N(k) = 0$ for every integer $k \geq 1$, and deduce that N is the zero polynomial.
- (c) Deduce that every $c_j = 0$, and hence a contradiction.

Solution.

- (a) Take $p = x^k$. Then $p(0) = 0$ for $k \geq 1$ and $p(0) = 1$ for $k = 0$, while

$$\langle x^k, w \rangle = \int_0^1 t^k \sum_{j=0}^d c_j t^j dt = \sum_{j=0}^d c_j \int_0^1 t^{k+j} dt = \sum_{j=0}^d \frac{c_j}{k+j+1}.$$

- (b) For an integer $k \geq 1$, multiply the identity of (a) by $\prod_{i=0}^d (k+i+1)$, a non-zero number. Each term $c_j/(k+j+1)$ becomes $c_j \prod_{i \neq j} (k+i+1)$, so the result is exactly $N(k) = 0$. Thus the polynomial N , of degree at most d , has the infinitely many distinct roots $1, 2, 3, \dots$. A non-zero polynomial of degree d has at most d distinct roots (Corollary 5.1.4), so $N = 0$.
- (c) Evaluate N at $s = -(j_0 + 1)$ for a fixed $j_0 \in \{0, \dots, d\}$. Every product $\prod_{i \neq j} (s+i+1)$ with $j \neq j_0$ contains the factor $s+j_0+1 = 0$ and so vanishes; the term $j = j_0$ contributes $c_{j_0} \prod_{i \neq j_0} (i-j_0)$, and that product is non-zero because the factors $i-j_0$ are non-zero integers. Since $N = 0$, we get $c_{j_0} = 0$. This holds for every j_0 , so $w = 0$. But then the case $k = 0$ of (a) reads $0 = 1$, a contradiction. Hence no Riesz vector exists.

The moral matches [Example \(A functional with no Riesz vector\)](#): point evaluation is a perfectly good linear functional, but on an infinite-dimensional space it is not represented by any vector, whichever of these two inner products we use.